

Summary

The Commonwealth Government manages billions of dollars' worth of water entitlements in the Murray–Darling Basin (MDB) with the aim of protecting and restoring water-dependent ecosystems. Managing this water efficiently is critical for maximising the health of ecosystems across the MDB for current and future generations.

Meeting environmental demands is challenging

- The challenge is that the timing and location of allocations received do not always align with the timing and location of environmental demands. This is due to the characteristics of water entitlements, which were primarily developed for irrigators, and the composition of environmental water portfolios.
- This misalignment can be addressed in part by using tools such as carryover and inter-valley transfers. However, these tools have limitations – carryover is subject to limits, evaporation losses and spills, while inter-valley transfers are subject to transfer restrictions and can only be used to move water between connected catchments. Allocation trade can be used to overcome some of these limitations.

There are three main types of allocation trade

- This paper explores three trading strategies that environmental water holders (EWHs) can use to better meet environmental demands:
 - selling in one catchment to buy in another catchment (trading water between catchments).
 - selling water in one year to buy in another (trading water over time).
 - selling water and using the proceeds to fund other inputs that also contribute towards environmental objectives (such as, controlling pests and weeds or removing barriers to fish passage).

Environmental water trade can help the environment

- Allocation trade provides EWHs with a tool that can be used to move water to when and where it is needed most or to fund other environmental inputs. For example, EWHs already use carryover to move water across years (including to meet early season demands, when there may be low or no new allocations). Yet, as mentioned above, this is subject to limits and losses which reduce the volume of water that can be accrued, especially when attempting to move water over several years. An alternative strategy is trade, in this case, selling water in one year to buy in another. The main advantage of trade is that revenue can be accrued over multiple years without the limits and losses associated with carryover. The revenue can then be used to purchase water when and where it is needed most.
- To illustrate, we developed a simple environmental water management model to explore the effectiveness of two strategies (carryover and trade) at meeting the demands of hypothetical wetlands in the Goulburn and Murrumbidgee that require inundation every five years. In the

modelling example, using the trade strategy results in average inundation volumes that are twice as high in the Goulburn and three times higher in the Murrumbidgee, where carryover rules are less flexible. This does not mean the trade strategy is superior to the carryover strategy in all circumstances, but it does show the potential for trade to unlock greater environmental benefits when used as part of a broader environmental management plan.

The impacts of environmental water trade on irrigators are largely neutral

- A potential concern is that environmental water trade could harm irrigators by increasing or decreasing water allocation prices. Our modelling of the Goulburn catchment shows that trades in the order of 25 per cent of Commonwealth Environmental Water Holder (CEWH) allocations received would not necessarily have a significant impact on water allocation prices, if spread throughout the year. That said, larger trades in the order of 100 per cent of CEWH allocations received could have a significant effect. The price impacts of trade are likely to differ across catchments depending on the market share of CEWH entitlements in each catchment and market characteristics of each catchment.
- While some larger environmental water trades could have a significant impact on water allocation prices, it does not necessarily follow that this would be harmful to irrigators. Take the case where EWHs are buying allocations. While large increases in water allocation prices are bad for irrigators looking to buy allocations, they are beneficial for irrigators looking to sell. Gains to sellers from higher prices will tend to offset losses to buyers. So, while significant increases in water allocation prices would be harmful for some irrigators, they would not necessarily be harmful to irrigators in aggregate.

There are practical ways to facilitate more environmental water trade

- To realise the potential benefits of environmental water trade for the environment, EWHs will need to change how they engage with the water allocation market. Other water market participants use water markets freely in pursuit of their objectives. EWHs should do the same.
- While challenges exist, there are practical steps to help make the most of the opportunities presented by environmental water trade, including:
 - Clarify the definition of ‘harm’ under the ‘good neighbour’ principle. In building its social licence, the CEWH has operated as a good neighbour, with a general principle of causing no harm to other water users. When considering price impacts from trading water, any negative impacts on some people tend to be offset by positive impacts on other people. That is, price impacts have distributional consequences, but they do not cause harm in aggregate, and are consistent with being a good neighbour.
 - Develop simple trading strategies, recognising that this will be an imperfect step in the right direction. In the long term, develop more sophisticated trading strategies, potentially drawing on mathematical optimisation methods.
 - Current legislation allows many but not all types of environmental water trade. In particular, there are constraints on the ability of the CEWH to sell water to fund other environmental activities that would also improve the health of water-dependent ecosystems. This raises the question of whether these types of trade should be allowed. This

is a complex long-term issue. It partly depends on whether we have the right balance between the multi-billion dollar environmental water portfolio and the resources available for other environmental activities. If not, and there is a significant imbalance, allowing the CEWH to sell water to fund other environmental activities is a potential solution. But there are theoretical drawbacks as well as practical challenges, and it is not the only potential solution. This report does not answer the question of whether legislative change is warranted. But we think it is worth investigating in the future, within the broader context described above.

Source: <https://www.agriculture.gov.au/abares/research-topics/water/environmental-water-trade-making-every-drop-count>